Discussion of “Trade Costs and Inflation”

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This paper

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2. Why is this paper important?
   - Monetary policy trade-offs: trade costs as cost-push shocks?
3. What do they do?
   - Measure trade costs.
   - Show empirically that trade costs change affect inflation:
     - if on final goods: one-time change in price level.
     - if on intermediate goods: persistent inflation.
   - Build a multi-country NK model to rationalize the findings.
   - Revisit 2018/19 US-China trade war and COVID-19 pandemic.

Thoughts and roadmap

- Great paper! been around for some time:

    *First version: February 2024. The views in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of any other person associated with the Federal Reserve System. We thank seminar participants at the University of Washington, the Federal Reserve Board, the Dallas Fed, the LAJCB Conference, CEBRA 2024 Annual Meetings, 2025 ASSA/AEA Annual Meetings, Brandeis University, West Coast Workshop in International Finance, the Mid-Atlantic Trade Workshop, and SCIEA Meeting at the NY Fed. We are particularly grateful to our discussants Matteo Cacciatore, Thuy Lan Nguyen, and Julian Di Giovanni, and to Francesco Bianchi, Daniel Xu, Carter Mix, Martin Uribe and Stephanie Schmitt-Grohé for comments and suggestions.

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- Let me try to do the following:
  1. Understanding mechanisms/related literature.
  2. Real GDP in open economies.
  3. Trade costs: iceberg costs or tariffs (time permitting).

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- Main result: higher trade costs
  - on intermediate goods $\longrightarrow$ small but persistent inflation
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- I like analytical results.
- Let me try to show your result in a stylized model.

“Simplified” model

- 2 countries. 2 goods.
- Country 1 produces good 1, country 2 produces good 2.
- Goal: what are the effects of trade costs on inflation in country 1?
- Inflation in country 1:

$$\pi_{1,1,t} = \omega_{11}^{C}\pi_{11,1,t} + \omega_{12}^{C}\pi_{12,1,t}$$

  with $\pi_{ij,s,t}$ inflation in country i of good j in currency s.
- If deviations from law of one price $\pi_{12,1,t} = \pi_{22,1,t} + \Delta\tau_{12,t}^{C}$.

$$\pi_{1,1,t} = \omega_{11}^{C}\pi_{11,1,t} + \omega_{12}^{C}\pi_{22,1,t} + \omega_{12}^{C}\Delta\tau_{12,t}^{C}$$

“Simplified” model ct’ed

- Inflation in country 1 under fully sticky wages is

$$\pi_{1,1,t} = (\boldsymbol{\omega}_1^C)' \boldsymbol{\pi}_t + \omega_{12}^C \Delta \tau_{12,t}^C$$

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- $\boldsymbol{\Psi}^{\text{Sticky}} = (\boldsymbol{I} - \hat{\chi}\tilde{\boldsymbol{\Omega}}^{\boldsymbol{M}})^{-1}\hat{\chi}$: Sticky Leontief inverse (my take)
  - function of price stickiness and trade linkages across countries.

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$$- \underbrace{(\boldsymbol{\omega}_{1}^{C})' \boldsymbol{\Psi}^{\text{Sticky}} (\boldsymbol{I} - \tilde{\boldsymbol{\Omega}}^{M}) \boldsymbol{P}_{t-1}}_{\text{Persistence}}$$

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$$- \underbrace{(\boldsymbol{\omega}_1^{C})'\boldsymbol{\Psi}^{\text{Sticky}}(\boldsymbol{I} - \tilde{\boldsymbol{\Omega}}^{M})\boldsymbol{P}_{t-1}}_{\text{Persistence}}$$
$$\underbrace{+(\boldsymbol{\omega}_1^{C})'\boldsymbol{\Psi}^{\text{Sticky}}(\hat{\boldsymbol{\chi}}^{-1} - \boldsymbol{I})\left[\beta E_t\boldsymbol{\pi}_{t+1} + \beta E_t\boldsymbol{\pi}_{t+1}^{\mathcal{E}} - \boldsymbol{\pi}_t^{\mathcal{E}}\right]}_{\text{Expectations + Exchange rate channels}}$$

- Multi-country version of multi-sector NK models with IO linkages.

Real GDP data vs. model

- Footnote 33 in the paper define real GDP as

$$GDP_{i,t} = \frac{P_{i,t}Y_{i,t} - P^M_{i,t}M_{i,t}}{P^C_{i,t}}$$

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- Real GDP in the data uses double-deflation.
- Important when studying open economies.

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- If divide nominal exports by import price (CPI) then $P_X X / P_M = M$.
  - Measure is not real GDP but imported quantity (consumption)!

CPI versus GDP deflator: Implications for Taylor Rule

- Taylor Rule in the paper

$$R_{i,t} = R_{i,t-1}^{\phi_r} \left((\pi_{i,t})^{\phi_\pi} \left(\frac{GDP_{i,t}}{GDP_{i,t}^{\text{flex}}} \right)^{\phi_y} \varepsilon_{i,t}^{r} \right)^{1-\phi_r}$$

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- Consumption or output gap?

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- Perhaps behind your almost identical results in Figure 8 2nd row?

Iceberg costs versus tariffs

- Iceberg costs are not rebated, tariffs are.
- Is this important? Yes! It affects the optimal monetary policy response
  - Optimal monetary policy response to tariffs is "expansionary".
    (e.g. Bianchi and Coulibaly, 2025; Werning, Guerrieri and Lorenzoni, 2025).
  - Why?
    - With distortions, optimal monetary policy want to get closer to first-best.
    - If permanent tariffs, new steady-state is distorted even if initial one is not.
    - Descentralized equilibrium is constrained inefficient with tariffs.
  - but I understand this is a positive paper!
  - My suggestion: no need to talk about tariffs when discussing the results.

Final thoughts

- Great paper. Important question. Well-written and executed.
- Pushed me to think even more seriously about these issues.

Thank you!

`asilvub.github.io`